Analysis I

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1 Logic, Sets and Numbers

1.1 Logic

Definition 1.1. *A **mathematical Statement** is a well-formulated assertion that is either strictly true or false.*

Example 1.2. *"Every prime number is odd." is a false statement.*

Definition 1.3. *A **Statement** that depends on one or multiple variables is called a **predicate**.*

Example 1.4.

$$P(x) \coloneqq x > 2$$

Definition 1.5. *The negation of a mathematical Statement A is defined as $\neg A \coloneqq$ A is false.*

Example 1.6. $A \coloneqq \forall x \in \mathbb{R} : x > 2$ $\neg A \coloneqq \exists x \in \mathbb{R} : x \leq 2$

Definition 1.7. *Quantifiers are short hand notation for making statements about predicates.*

$$\forall x \in \mathbb{R} : P(x)$$

$$\exists x \in \mathbb{R} : P(x)$$

$$\exists! x \in \mathbb{R} : P(x)$$

$$\nexists x \in \mathbb{R} : P(x)$$

Expresses that for all / for at least one / for exactly one or for none x in $\mathbb{R}$ satisfies the predicate.

Important 1.8. *The oderder of quantifiers matters. $\forall x \exists y$ is not equivalent to $\exists y \forall x$.*

Definition 1.9. *Basic logical connectives are defined as:*

$$A \wedge B \coloneqq A \text{ and } B \text{ are true}$$

$$A \vee B \coloneqq A \text{ or } B \text{ is true}$$

$$A \implies B \coloneqq A \text{ implies } B \text{ is true}$$

$$A \iff B \coloneqq A \text{ is equivalent to } B \text{ is true}$$

Remark 1.10. *Observe that equivalence holds iff.* $A \implies B \wedge B \implies A$

Lemma 1.11. *From the above definition it follows that:*

$$\neg A \vee B \iff A \implies B$$

This can be shown by evaluating the truth table for both statements.

Theorem 1.12. *The following equivalences hold:*

$$\neg(\forall x P(x)) \iff \exists x \neg P(x)$$

$$\neg(\exists x P(x)) \iff \forall x \neg P(x)$$

aswell as

$$\forall x(P(x) \wedge Q(x)) \iff (\forall x P(x)) \wedge (\forall x Q(x))$$

$$\exists x(P(x) \vee Q(x)) \iff (\exists x P(x)) \vee (\exists x Q(x))$$

1.2 Sets

Definition 1.13. *A* ***set*** *is a collection of elements, that are listed or characterized by a certain property. The empty set is denoted by* $\emptyset$ *and is the only set that contains no elements.*

Definition 1.14. *An element* x *is contained in a set* A *iff.* $x \in A$ *else it is not contained in* A *denoted by* $x \notin A$.

Definition 1.15. *Let* A *and* B *be sets.* A *is a* ***subset*** *of* B *denoted by* $A \subseteq B$ *iff.*

$$\forall x(x \in A \implies x \in B)$$

Definition 1.16. A *is a* ***proper subset*** *of* B *denoted by* $A \subset B$ *iff.*

$$A \subseteq B \wedge A \neq B$$

Definition 1.17. *Let* A *and* B *be sets.*

- *The* ***union*** *of* A *and* B *is defined as* $A \cup B \coloneqq \{x \mid x \in A \vee x \in B\}$.

- *The **intersection** of A and B is defined as $A \cap B := \{x \mid x \in A \wedge x \in B\}$.*
- *The **difference** of A and B is defined as $A \setminus B := \{x \mid x \in A \wedge x \notin B\}$.*
- *The **Cartesian product** of A and B is defined as $A \times B := \{(a, b) \mid a \in A \wedge b \in B\}$.*

1.3 Numbers

The basic and often used sets of numbers are:

- $\mathbb{N}$: The set of natural numbers 1, 2, 3, 4, ...
- $\mathbb{Z}$: The set of integers ... -2, -1, 0, 1, 2, ...
- $\mathbb{Q}$: The set of rational numbers $\frac{p}{q}$ where $p, q \in \mathbb{Z}$
- $\mathbb{R}$: The set of real numbers

Remark 1.18. *Some extensions of the real numbers are $\mathbb{R}_0^+/\mathbb{R}^+$ and $\mathbb{R}_0^-/\mathbb{R}^-$ which define the intervalls $[0, \infty)/(0, \infty)$ and $(-\infty, 0]/(-\infty, 0)$.*

Definition 1.19. *Continuous subsets are called **intervalls.** An intervall is called*

- ***open** if it does not contain its endpoints $I = (a, b) := \{x \mid a < x < b\}$*
- ***closed** if it contains its endpoints $I = [a, b] := \{x \mid a \leq x \leq b\}$*
- ***half open** if it contains one endpoint and not the other $I = [a, b) := \{x \mid a \leq x < b\}$*

Definition 1.20. *An intervall is **bounded** if it is bounded from above and below hence a and b are finite.*

Remark 1.21. *A bounded and closed intervall is often refered to as **compact**.*

Definition 1.22. *An upper (lower) bound of a subset $X \subset \mathbb{R}$ is an element $y \in \mathbb{R}$ such that:*

$$\forall x \in X : x \leq (\geq) y$$

The maximum (minimum) of a subset $X \subset \mathbb{R}$ is the element $x_0 \in X$ such that:

$$\forall x \in X : x \leq (\geq) x_0$$

Definition 1.23. *The* ***Supremum*** $sup(X)$ *of a subset* $X \subset \mathbb{R}$ *is the least upper bound of* X*. and can be characterized by:*

$$\forall x \in X : x \leq sup(X) \wedge \forall \epsilon > 0 : \exists x \in X : x > sup(X) - \epsilon$$

Definition 1.24. *The* ***Infimum*** $inf(X)$ *of a subset* $X \subset \mathbb{R}$ *is the greatest lower bound of* X*. and can be characterized by:*

$$\forall x \in X : x \geq inf(X) \wedge \forall \epsilon > 0 : \exists x \in X : x < inf(X) + \epsilon$$

Example 1.25. *Let* $X = [-2, 1)$ *we have a minimum of* $x_0 = -1$ *but no maximum. Observe that it is bounded from above by* 1 *and* 2 *or any number larger than* 1 *but only* 1 *is the least upper bound (Supremum).*

Theorem 1.26. *The following holds:*

- *The* ***maximum and minimum are unique*** *aslong as they exist.*
- *Every* ***non empty, bounded from above (below)*** *subset of* $\mathbb{R}$ *has a* ***maximum (minimum)****.*

Definition 1.27. *A set* X *is said to be (order)complete iff for every non empty subsets* $A, B \subseteq X$

$$\forall a \in A \forall b \in B : a \leq b$$

$$\implies \exists c \in X \quad s.t. \quad \forall a \in A : a \leq c \wedge \forall b \in B : c \leq b$$

Example 1.28. $\mathbb{Q}$ *is not order complete. Let* $A = \{x \in \mathbb{Q} \mid x^2 \leq 2\}$ *and* $B = \{x \in \mathbb{Q} \mid x^2 \geq 2\}$*. Then* $\forall a \in A \forall b \in B : a \leq b$ *but there is no* $c \in \mathbb{Q}$ *such that* $c^2 = 2$*.*

Theorem 1.29. *The Archimedian Principle says that*

- *For* $x \in \mathbb{R}$ *and* $y > 0$ *there exists an element* $n \in \mathbb{N}$ *such that* $n * y > x$*.*
- *For every* $\epsilon > 0$ *there exists an element* $n \in \mathbb{N}$ *such that* $\frac{1}{n} < \epsilon$*.*

(This follows from the order completenes of $\mathbb{R}$*.)*

Theorem 1.30. *The Young Inequality states that for all* $\epsilon > 0$ *and all* $x, y \in \mathbb{R}$*:*

$$2|xy| \leq \epsilon x^2 + \frac{1}{\epsilon} y^2$$

Proof.

$$\left(\sqrt{\epsilon}x - \frac{1}{\sqrt{\epsilon}}y\right)^2 \geq 0 \implies \epsilon x^2 - 2xy + \frac{1}{\epsilon}y^2 \geq 0 \implies \epsilon x^2 + \frac{1}{\epsilon}y^2 \geq 2xy$$

□

Theorem 1.31. *The real numbers $\mathbb{R}$ satisfy axiomatic properties for addition and multiplication (forming a field):*

- *Associativity:* $x + (y + z) = (x + y) + z$ *and* $x \cdot (y \cdot z) = (x \cdot y) \cdot z$
- *Commutativity:* $x + y = y + x$ *and* $x \cdot y = y \cdot x$
- *Neutral elements:* $\exists 0, 1 \in \mathbb{R}$
- *Inverse elements:* $-x$ *and* x^{-1} *(for* $x \neq 0$*)*
- *Distributivity:* $x \cdot (y + z) = x \cdot y + x \cdot z$

$\mathbb{R}$ *has order axioms: reflexivity, transitivity, antisymmetry, totality, making it an* ***ordered and order-complete field.***

1.3.1 Vectors and Vector Spaces

Definition 1.32. *For $x, y \in \mathbb{R}^n$, the standard scalar product is defined as:*

$$x \cdot y = \langle x, y \rangle := \sum_{i=1}^{n} x_i y_i$$

The Euclidean norm is defined as:

$$||x|| := \sqrt{\sum_{i=1}^{n} x_i^2}$$

The Euclidean distance between x and y is:

$$d(x, y) := \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

Theorem 1.33. *The triangle inequality holds for all $x, y, z \in \mathbb{R}^n$:*

$$||x - z|| \leq ||x - y|| + ||y - z||$$

1.3.2 Complex Numbers

Definition 1.34. *The* ***complex numbers*** $\mathbb{C}$ *are numbers of the form:*

$$z = a + ib \quad \text{where } a, b \in \mathbb{R}$$

a is the real part $Re(z)$, and b is the imaginary part $Im(z)$. The imaginary unit i has the property $i^2 = -1$.

Definition 1.35. *Arithmetic of complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$:*

- *Addition:* $z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$
- *Multiplication:* $z_1 \cdot z_2 = (x_1x_2 - y_1y_2) + i(x_1y_2 + y_1x_2)$

Definition 1.36. *The **complex conjugate** of $z = x + iy$ is $\bar{z} = x - iy$.*

Important 1.37. *Division is calculated by expanding with the conjugate of the denominator:*

$$\frac{z}{w} = \frac{z \cdot \bar{w}}{w \cdot \bar{w}} = \frac{z \cdot \bar{w}}{|w|^2}$$

Remark 1.38. *Some usfull properties of the conjugate:*

- $z \cdot \bar{z} = |z|^2$
- $z + \bar{z} = 2Re(z)$ *and* $z - \bar{z} = 2iIm(z)$
- $\overline{z + w} = \bar{z} + \bar{w}$ *and* $\overline{z \cdot w} = \bar{z} \cdot \bar{w}$

Definition 1.39. *Geometrically, complex numbers can be represented in the Gaussian plane. The **absolute value** of $z = a + ib$ is its **distance from the origin**:*

$$|z| = \sqrt{a^2 + b^2}$$

The distance between z_1 and z_2 is $d = |z_1 - z_2|$.

Remark 1.40. *Observe that the triangle inequality holds in $\mathbb{C}$: $|z + w| \leq |z| + |w|$.*

Theorem 1.41. *The Euler's formula states that for $t \in \mathbb{R}$:*

$$e^{it} = \cos(t) + i\sin(t)$$

This allows writing trigonometric functions using complex exponentials:

$$\cos(t) = \frac{e^{it} + e^{-it}}{2}, \quad \sin(t) = \frac{e^{it} - e^{-it}}{2i}$$

Remark 1.42. *The* ***Intuition*** *behind eulers formula is we can use Taylor series to show that:*

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

Using $x = it$ we get:

$$e^{it} = \sum_{n=0}^{\infty} \frac{(it)^n}{n!} = \cdots = \cos(t) + i\sin(t)$$

Eulers formula leads us to a new representation of complex numbers: the polarcoordinates.

Definition 1.43. *For $z \in \mathbb{C}$, the polar-coordinates are defined as:*

$$z = r(\cos(t) + i\sin(t)) = re^{it}$$

where $r = |z|$ is the absolute value and $t = \arg(z)$ is the argument.

Multiplication and division in polar-coordinates is much easier than in cartesian coordinates:

$$z_1 \cdot z_2 = r_1 e^{it_1} \cdot r_2 e^{it_2} = r_1 r_2 e^{i(t_1+t_2)}$$

$$\frac{z_1}{z_2} = \frac{r_1 e^{it_1}}{r_2 e^{it_2}} = \frac{r_1}{r_2} e^{i(t_1-t_2)}$$

So is exponentiation or taking roots:

$$z^n = (re^{it})^n = r^n e^{int}$$

$$\sqrt[n]{z} = \sqrt[n]{r} e^{i\frac{t+2\pi k}{n}}, \quad k \in \{0, 1, \ldots, n-1\}$$

Definition 1.44. *The n-th roots of unity are the solutions to $z^n = 1$, which are given by:*

$$z_k = e^{i\frac{2\pi k}{n}}, \quad k \in \{0, 1, \ldots, n-1\}$$

Theorem 1.45. *The sum of the roots of unit is always zero.*

$$\sum_{k=0}^{n-1} e^{i\frac{2\pi k}{n}} = 0$$

Theorem 1.46. *The fundamental theorem of algebra states that every polynomial of degree $n \geq 1$ has exactly n roots in $\mathbb{C}$ (counting multiplicity). These roots appear in konplex-conjugate pairs.*

2 Sequences and Series

2.1 Sequences

Definition 2.1. *A* ***sequence*** *(Folge)* $(a_n)_{n\in\mathbb{N}_0}$ *is an infinite list of numbers, mathematically a mapping* $f : \mathbb{N}_0 \to X$*. It can be given explicitly or recursively as*

$$a_n = f(n) \quad \text{or} \quad a_n = g(a_{n-1}, a_{n-2}, \dots)$$

Example 2.2. *A famous example is the Fibonacci sequence defined by* $a_0 = 0, a_1 = 1$ *and* $a_n = a_{n-1} + a_{n-2}$ *for* $n \geq 2$*. The first few terms are* $0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots$

Definition 2.3. *A sequence* (a_n) *is called* ***convergent*** *with limit* $L \in \mathbb{R}$ *if:*

$$\forall \epsilon > 0 \exists N > 0 \quad s.t. \quad \forall n > N : |a_n - L| < \epsilon$$

Written as $\lim_{n\to\infty} a_n = L$*.If no such* L *exists, the sequence is* ***divergent****. Sequences that diverge to* $\pm\infty$ *are a special case of divergent sequences.*

Theorem 2.4. *The limit of a sequence is unique aslong as it exists.*

Proof. Assume (a_n) converges to L_1 and L_2. Then for any $\epsilon > 0$ there exists N_1 and N_2 such that:

$$|a_n - L_1| < \epsilon \quad \forall n > N_1$$
$$|a_n - L_2| < \epsilon \quad \forall n > N_2$$

Then for any $n > \max(N_1, N_2)$:

$$|L_1 - L_2| = |L_1 - a_n + a_n - L_2| \leq |L_1 - a_n| + |a_n - L_2| < 2\epsilon$$

Since this holds for any $\epsilon > 0$, we must have $L_1 = L_2$. □

Theorem 2.5. *A subsequence* $(a_{n_k})_{k\in\mathbb{N}_0}$ *of a convergent sequence converges to the same limit* L*.*

Definition 2.6. *An* ***accumulation point*** *(Häufungspunkt)* $A \in \mathbb{R}$ *of a sequence fulfills:*

$$\forall \epsilon > 0 \forall N \in \mathbb{N}_0 \exists n \geq N \text{ such that } |a_n - A| < \epsilon$$

Theorem 2.7. *A is an accumulation point iff there exists a convergent subsequence with limit* A*.*

Proof. Let (a_n) be a sequence and A an accumulation point. We can easily construct a subsequence (a_{n_k}) that converges to A by selecting n_k such that $|a_{n_k} - A| < 1/k$ this is given to exist by the definition of an accumulation point. Now let (a_{n_k}) be a convergent subsequence with limit A then for any $\epsilon > 0$ there exists K such that $|a_{n_k} - A| < \epsilon$ for all $k > K$ and since $n_k \to \infty$ as $k \to \infty$ by definition it follows that A is an accumulation point. □

Theorem 2.8. *Every convergent sequence has exactly one accumulation point (its limit).*

Proof. For A to be an accumulation point of a_n we need that there are indefinitely many a_n that approach A. If a_n converges to L then we can intuitively say that that A has to be L. □

Theorem 2.9. *Limit arithmetic: For* $\lim a_n = K$, $\lim b_n = L$ *and* $C \in \mathbb{R}$*:*

- $\lim(a_n \pm b_n) = K \pm L$
- $\lim(C \cdot a_n \cdot b_n) = C \cdot K \cdot L$
- $\lim(a_n/b_n) = K/L$ *(if* $L \neq 0$ *and* $b_n \neq 0$*)*

Theorem 2.10. ***Sandwich-Theorem:*** *If* $a_n \leq b_n \leq c_n$ *for* $n \geq N_0$*, and* $\lim a_n = \lim c_n = L$*, then* $\lim b_n = L$*.*

Proof. Let $\lim a_n = \lim c_n = L$ and $\lim b_n = M$. From the above arithmetic we can sa that $a_n \leq b_n \leq c_n$ implies $L \leq M \leq L$ so we have that b_n is squeezed between a_n and c_n and thus must converge to $M = L$. □

Remark 2.11. *The sandwich theorem is often used to find upper and lower bounds for a sequence that converge to the same limit and therfore conclude that the original sequence does too.*

Remark 2.12. *Important limits:*

- $\lim_{n\to\infty} \frac{\ln n}{n} = 0$
- $\lim_{n\to\infty} n^{1/n} = 1$
- $\lim_{n\to\infty} x^{1/n} = 1 \quad (x > 0)$
- $\lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n = e^x$
- $\lim_{n\to\infty} \frac{x^n}{n!} = 0$

Definition 2.13. *A sequence is*

- ***monotonically increasing (decreasing)*** *if* $a_n \leq (\geq) a_m$ *for* $m > n$.
- ***upper (lower) bounded*** *if there exists* M *such that* $|a_n| \leq (\geq) M \quad \forall n \in \mathbb{N}_0$.

Theorem 2.14. ***Bolzano-Weierstrass Theorem:*** *Every bounded sequence has an accumulation point.*

Theorem 2.15. ***Monotone Convergence Theorem:*** *Every bounded monotonic sequence converges to its supremum (infimum).*

Definition 2.16. *The limit of the supremum is called* ***limes superior*** *and denoted as:*

$$\limsup_{n\to\infty} a_n = \lim_{n\to\infty} \sup_{k\geq n} a_k$$

The limit of the infimum is called ***limes inferior*** *and denoted as:*

$$\liminf_{n\to\infty} a_n = \lim_{n\to\infty} \inf_{k\geq n} a_k$$

Definition 2.17. *A sequence is a* ***Cauchy sequence*** *iff*

$$\forall \epsilon > 0 \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N : |a_n - a_m| < \epsilon$$

Theorem 2.18. *A sequence of real numbers converges iff it is a cauchy sequence.*

Proof. $\Longrightarrow$: Let (a_n) converge to L. For any $\epsilon > 0$, there exists N such that $\forall n \geq N$: $|a_n - L| < \epsilon/2$. Then for any $m, n \geq N$:

$$|a_n - a_m| = |a_n - L + L - a_m| \leq |a_n - L| + |L - a_m| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Thus, (a_n) is a Cauchy sequence.

$\Longleftarrow$: Let (a_n) be a Cauchy sequence. First, (a_n) is bounded: choose $\epsilon = 1$, then there exists N such that $|a_n - a_N| < 1$ for all $n \geq N$. Thus $|a_n| < |a_N| + 1$ for $n \geq N$, meaning (a_n) is bounded by $M = \max(|a_0|, \ldots, |a_{N-1}|, |a_N| + 1)$. By the Bolzano-Weierstrass theorem, the bounded sequence (a_n) has an accumulation point L, so there exists a subsequence (a_{n_k}) converging to L. For any $\epsilon > 0$, since (a_n) is Cauchy, there exists N' such that $\forall m, n \geq N' : |a_n - a_m| < \epsilon/2$. Since $a_{n_k} \to L$, choose a k such that $n_k \geq N'$ and $|a_{n_k} - L| < \epsilon/2$. Then for any $n \geq N'$:

$$|a_n - L| \leq |a_n - a_{n_k}| + |a_{n_k} - L| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Thus, the sequence converges to L. $\square$

2.2 Series

Definition 2.19. *A* ***series*** *(Reihe) is an infinite sum* $\sum_{i=0}^{\infty} a_i$ *of the terms of a sequence* (a_n)*.*

Remark 2.20. *A series is said to converge iff the sequence of partialsums converges.*

Theorem 2.21. ***Necessary condition for convergence:*** *for* $\sum_{i=0}^{\infty} a_i$ *to converge we need* $\lim_{n\to\infty} a_n = 0$*.*

Remark 2.22. *Some examples of series are:*

- ***geometric series****:* $\sum_{n=0}^{\infty} x^n$ *converges to* $\frac{1}{1-x}$ *iff* $|x| < 1$
- ***harmonic series****:* $\sum_{n=1}^{\infty} \frac{1}{n}$ *converges for* $s > 1$

Theorem 2.23. *For* ***series with non-negative terms*** $a_n \geq 0$*, the sequence of partial sums is monotonically increasing.* ***It converges iff it is bounded.***

Theorem 2.24. ***Comparison test*** *(Vergleichskriterium): If* $c_n \leq b_n \leq a_n$ *for* $n \geq N$*:*

- *If* $\sum a_n$ *converges, then* $\sum b_n$ *converges (****Majorantenkriterium****).*
- *If* $\sum c_n$ *diverges, then* $\sum b_n$ *diverges (****Minorantenkriterium****).*

Example 2.25.

$$\sum_{n=1}^{\infty} \frac{1}{n!} = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \dots$$

We can compare this series to the geometric series

$$1 + \sum_{n=1}^{\infty} (\frac{1}{2})^n = 1 + 2 = 3$$

Therfore accordint to the Majorantenkriterium the series $\sum_{n=1}^{\infty} \frac{1}{n!}$ *converges.*

Definition 2.26. *A series is* ***absolutely convergent*** *if* $\sum |a_n|$ *converges. Absolute convergence implies normal convergence. A* ***conditionally convergent*** *series converges, but not absolutely.*

Theorem 2.27. ***Riemann's rearrangement theorem:*** *Let $\sum a_n$ be a conditionally convergent series. Then for any $x \in \mathbb{R}$, there exists a rearrangement of the series that converges to x.*

Theorem 2.28. ***Leibniz Criterion****: For an alternating series $\sum(-1)^n a_n$ with a monotonically decreasing null sequence (a_n), the series converges.*

Proof. Let (a_n) be a non-negative, monotonically decreasing sequence converging to 0. Let $s_n = \sum_{k=0}^{n}(-1)^k a_k$ be the n-th partial sum. For the even partial sums (s_{2n}), we have

$$s_{2n+2} - s_{2n} = -a_{2n+1} + a_{2n+2} \leq 0$$

since $a_{2n+2} \leq a_{2n+1}$. Thus, (s_{2n}) is monotonically decreasing. Similarly for the odd partial sums (s_{2n+1}), we have

$$s_{2n+3} - s_{2n+1} = a_{2n+2} - a_{2n+3} \geq 0$$

meaning (s_{2n+1}) is monotonically increasing. Note that

$$s_{2n+1} = s_{2n} - a_{2n+1} \leq s_{2n} \leq s_0 = a_0$$

This limits both bounds: $0 \leq s_{2n+1} \leq s_{2n} \leq a_0$. By the Monotone Convergence Theorem, both bounded monotonic subsequences converge: $\lim_{n\to\infty} s_{2n} = L_1$ and $\lim_{n\to\infty} s_{2n+1} = L_2$. Finally,

$$L_1 - L_2 = \lim_{n\to\infty}(s_{2n} - s_{2n+1}) = \lim_{n\to\infty} a_{2n+1} = 0$$

So $L_1 = L_2 = L$. Since both the even and odd subsequences converge to the same limit L, the entire sequence of partial sums (s_n) converges to L, thus the series converges. $\square$

Theorem 2.29. ***Cauchy Criterion:*** *The series $\sum a_n$ converges iff*

$$\forall \epsilon > 0 \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N : |\sum_{k=n}^{m} a_k| < \epsilon$$

Theorem 2.30. *More criterions for convergence:*

- ***Root Criterion*** *(Wurzelkriterium): Let $\rho = \limsup \sqrt[n]{|a_n|}$. If $\rho < 1$, abs conv; if $\rho > 1$, div.*
- ***Ratio Criterion*** *(Quotientenkriterium): Let $\rho = \lim \left|\frac{a_{n+1}}{a_n}\right|$. If $\rho < 1$, abs conv; if $\rho > 1$, div.*

Example 2.31.

$$\sum_{n=1}^{\infty} n^4 e^{-n}$$

$$\rho = \lim_{n\to\infty} \frac{(n+1)^4 e^{-(n+1)}}{n^4 e^{-n}} = \lim_{n\to\infty} \left(\frac{n+1}{n}\right)^4 e^{-1} = e^{-1} < 1$$

Therfore the series converges.

Definition 2.32. *A* ***power series*** *(Potenzreihe) centered at a has the form*

$$\sum_{k=0}^{\infty} c_k (x-a)^k$$

Theorem 2.33. *Every power series has a* ***convergence radius*** *R, with the* ***interval of convergence*** *$(a-R, a+R)$. R is computed by the formula $R = \frac{1}{\limsup |c_n|^{1/n}}$.*

Example 2.34.

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n} x^n$$

$$\rho = \limsup \frac{1}{n^{1/n}} = 1 \implies R = 1$$

Therfore the series converges for

$$x \in (-1, 1)$$

.

Remark 2.35. *For $x = 1$ we have the harmonic series, which diverges. For $x = -1$ we have the alternating harmonic series, which converges.*

3 Functions

Definition 3.1. *A* ***function*** $f : D \to C$ *is a map from a domian* D *to a codomain* C *s.t. for each* $x \in D$ *there is exactly one* $y \in C$ *st.* $f(x) = y$.

Remark 3.2. *The input is called the* ***independent variable*** *(argument) while the output is called the* ***dependent variable****.*

Definition 3.3. *A function* f *is called*

- ***injective*** *iff* $\forall x_1, x_2 \in A : f(x_1) = f(x_2) \implies x_1 = x_2$.
- ***surjective*** *iff* $\forall y \in B \exists x \in A : f(x) = y$.
- ***bijective*** *iff* f *is injective and surjective.*

Example 3.4. *Let* $f : \mathbb{R} \to \mathbb{R}$ *with* $f(x) = x^2$. *Observe that* f *is not surjective as no output is negative, nor is it injective as for ex.* $f(1) = f(-1) = 1$. *But* $g : \mathbb{R}_0^+ \to \mathbb{R}_0^+$ *with* $g(x) = x^2$ *is both injective and surjective.*

Definition 3.5. *The* ***restriction*** *of* f *under* $D' \subset \mathbb{D}$ *is defined as* $f \restriction_{D'} : D' \to C, x \mapsto f(x) \quad \forall x \in \mathbb{D}$.

Remark 3.6. *Note that* $f \restriction_{D'}$ *and* f *are two diffrent functions.*

Definition 3.7. *Let* $f : D \to C$ *and* $g : C \to B$ *be functions then*

$$g \circ f : D \to B \quad \text{with} \quad g \circ f(x) = g(f(x)) \quad \forall x \in D$$

is called the ***composition*** *of* f *and* g.

Definition 3.8. *The* ***identity function*** $id_D : D \to D$ *is defined as* $id_D(x) = x \quad \forall x \in D$.

Definition 3.9. *If a function* $f : D \to C$ *is bijective then there exists a unique function* $f^{-1} : C \to D$ *such that*

$$f(f^{-1}(y)) = id_C = y \quad \forall y \in C$$

and

$$f^{-1}(f(x)) = id_D = x \quad \forall x \in D$$

f^{-1} *is called the* ***inverse function*** *of* f.

Definition 3.10. *A function* f *is called* ***bouned*** *if there exists* $M \in \mathbb{R}$ *such that* $|f(x)| \leq (\geq) M$ *for all* $x \in D$.

Definition 3.11. *A function* $f : \mathbb{R} \to \mathbb{R}$ *is called*

- ***(strictly) monotonically increasing*** *iff* $\forall x_1, x_2 \in \mathbb{R} : x_1 < x_2 \implies f(x_1) < f(x_2)$
- ***(strictly) monotonically decreasing*** *iff* $\forall x_1, x_2 \in \mathbb{R} : x_1 < x_2 \implies f(x_1) > f(x_2)$
- ***(strictly) monotonic*** *iff* f *is either (strictly) monotonically increasing or (strictly) monotonically decreasing*

Theorem 3.12. *Every strictly monotonic function is injective.*

Proof. Let f be strictly monotonic (wlog increasing) and assume it is not injective then there exists $x_1 \neq x_2$ such tat $f(x_1) = f(x_2)$. Wlog assume $x_1 < x_2$ then $f(x_1) < f(x_2)$ which contradicts the assumption that $f(x_1) = f(x_2)$. □

Definition 3.13. *A function* $f : \mathbb{R} \to \mathbb{R}$ *is called*

- ***even*** *iff* $\forall x \in \mathbb{R} : f(-x) = f(x)$
- ***odd*** *iff* $\forall x \in \mathbb{R} : f(-x) = -f(x)$

Example 3.14. *The function* $f : \mathbb{R} \to \mathbb{R}$ *with* $f(x) = x^3$ *is odd as* $f(-x) = (-x)^3 = -x^3 = -f(x)$ *while the function* $g : \mathbb{R} \to \mathbb{R}$ *with* $g(x) = x^2$ *is even as* $g(-x) = (-x)^2 = x^2 = g(x)$.

Remark 3.15. *Observe that functions do not have to be even or odd.*

Definition 3.16. *The* ***graph*** *of a function* $f : D \to C$ *is the set of all ordered pairs* $(x, f(x))$ *for* $x \in D$. *It is* ***konvex*** *if it lies below the line segment connecting any two points on the graph. Similarly it is called* ***konkav*** *if it lies above the line segment connecting any two points on the graph.*

3.1 Limits in the context of functions

Definition 3.17. *Let $f : \mathbb{D} \to \mathbb{R}$ be a function satisfying $\mathbb{D} \cap (x_0 - \delta, x_0 + \delta) \neq \emptyset$ for all $\delta > 0$ and $x_0 \in \mathbb{R}$. We say that $\lim_{x \to x_0} f(x) = L$ iff*

$$\forall \varepsilon > 0 \exists \delta > 0 s.t.$$

$$x \in \mathbb{D} \cap (x_0 - \delta, x_0 + \delta) \implies |f(x) - L| < \varepsilon$$

The limit L is unique.

Remark 3.18. *The intuition behind the definition of the limit is that for any given distance $\varepsilon > 0$ around L, we can find a $\delta > 0$ such that for all $x \in \mathbb{D}$ that are at least δ close to x_0, we have $|f(x) - L| < \varepsilon$. So for any boound on the output (ε) there exists a bound on the input (δ) such that all elements in the input bound $(x_0 - \delta, x_0 + \delta)$ are mapped to the output bound $(L - \varepsilon, L + \varepsilon)$.*

Definition 3.19. *A function $f : \mathbb{D} \to \mathbb{R}$ has in x_0 a*

- ***right-sided limit*** $\lim_{x \searrow x_0} f(x) = \lim_{x \to x_0^+} f(x) = L$ *iff*

$$\forall \varepsilon > 0 \exists \delta > 0 \quad s.t. \quad \forall x \in \mathbb{D}(x_0 < x < x_0 + \delta \implies |f(x) - L| < \varepsilon)$$

- ***left-sided limit*** $\lim_{x \nearrow x_0} f(x) = L$ *iff*

$$\forall \varepsilon > 0 \exists \delta > 0 \quad s.t. \quad \forall x \in \mathbb{D}(x_0 - \delta < x < x_0 \implies |f(x) - L| < \varepsilon)$$

Definition 3.20. *A function $f : \mathbb{D} \to \mathbb{R}$ has a*

- ***limit for x approaching infinity*** $\lim_{x \to \infty} f(x) = L$ *iff*

$$\forall \varepsilon > 0 \exists M > 0 \quad s.t. \quad \forall x \in \mathbb{D}(x > M \implies |f(x) - L| < \varepsilon)$$

- ***limit for x approaching negative infinity*** $\lim_{x \to -\infty} f(x) = L$ *iff*

$$\forall \varepsilon > 0 \exists M > 0 \quad s.t. \quad \forall x \in \mathbb{D}(x < M \implies |f(x) - L| < \varepsilon)$$

Definition 3.21. *A function $f : \mathbb{D} \to \mathbb{R}$ has in x_0 a*

- ***improper limit approaching infinity*** $\lim_{x \to x_0} f(x) = \infty$ *iff*

$$\forall N > 0 \exists \delta > 0 \quad s.t. \quad \forall x \in \mathbb{D}(0 < |x - x_0| < \delta \implies f(x) > N)$$

- ***improper limit approaching negative infinity*** $\lim_{x \to x_0} f(x) = -\infty$ *iff*

$$\forall N > 0 \exists \delta > 0 \quad s.t. \quad \forall x \in \mathbb{D}(0 < |x - x_0| < \delta \implies f(x) < -N)$$

Theorem 3.22. *Let* $A \in \mathbb{R}$ *and assume* $\lim_{x \to x_0} f(x) = K (\neq \pm\infty)$ *and* $\lim_{x \to x_0} g(x) = L (\neq \pm\infty)$.

- $\lim_{x \to x_0}(f(x) + g(x)) = K + L$
- $\lim_{x \to x_0}(f(x) - g(x)) = K - L$
- $\lim_{x \to x_0}(f(x)g(x)) = KL$
- $\lim_{x \to x_0}(f(x)/g(x)) = K/L$
- $\lim_{x \to x_0}(Af(x)) = AK$
- $\lim_{x \to x_0}(f(x)/g(x)) = K/L$ *if* $L \neq 0$

4 Continuity of functions

Definition 4.1. *A function $f : D \to \mathbb{R}$ is called* ***continous*** *at x_0 if*

$$\forall \epsilon > 0 \exists \delta > 0 \text{ such that } \forall x \in I(\mid x - x_0 \mid < \delta \implies \mid f(x) - f(c) \mid > \epsilon)$$

The function is called continuous if it is continuous at every point in its domain.

Definition 4.2. *Let $f : D \to \mathbb{R}$ be a function. Then f is called* ***left-continuous*** *at x_0 if*

$$\forall \epsilon > 0 \exists \delta > 0 \text{ such that } \forall x \in I(\mid x - x_0 \mid < \delta \implies \mid f(x) - f(c) \mid > \epsilon)$$

Theorem 4.3. *Let $f : D \subset \mathbb{D}(f) \subset \mathbb{R}$ be a function and $x_0 \in D$. f is continuous at x_0 iff for every sequence $(x_n)_{n\in\mathbb{N}} \subset D$ with $\lim_{n\to\infty} x_n = x_0$ it holds that $\lim_{n\to\infty} f(x_n) = f(x_0)$.*

Remark 4.4. *Continuity can also be viewed in terms of intervals, where a function f is continuous iff for every open interval $I \subset \mathbb{R}$ it holds that $f^{-1}(I)$ is open.*

Lemma 4.5. *Let f, g be continuous functions, the follwing holds:*

- *$f + g$ is continuous*
- *$f - g$ is continuous*
- *$f \cdot g$ is continuous*
- *$c \cdot f$ is continuous for all $c \in \mathbb{R}$*
- *$\frac{f}{g}$ is continuous if $g(x) \neq 0$*
- *$f \circ g$ is continuous*

Proof. We prove that $f \circ g$ is continuous if f and g are. Let g be continuous at a point c, and let f be continuous at the point $g(c)$.

By the limit definition of continuity, since g is continuous at c, we know:

$$\lim_{x\to c} g(x) = g(c)$$

Similarly, since f is continuous at $g(c)$, for any variable y approaching $g(c)$, we know:

$$\lim_{y\to g(c)} f(y) = f(g(c))$$

To find the limit of the composite function $f(g(x))$ as $x \to c$, we substitute $g(x)$ for our input y. Because f is continuous, we are allowed to pass the limit directly inside the function f:

$$\lim_{x \to c} f(g(x)) = f\left(\lim_{x \to c} g(x)\right)$$

Now, we just substitute the known limit of $g(x)$:

$$f\left(\lim_{x \to c} g(x)\right) = f(g(c))$$

Therefore, $\lim_{x \to c}(f \circ g)(x) = (f \circ g)(c)$. This perfectly satisfies the definition of continuity, proving that $f \circ g$ is continuous at c $\square$

Theorem 4.6. *Let I be an interval and let $f : I \to J$ be a continuous and bijective function. Then J is an interval too and the inverse function $f^{-1} : J \to I$ is continuous.*

Theorem 4.7. *Let $f : D \to \mathbb{R}$ be a continuous function. Let $c \in [f(a), f(b)]$ then there exists a $x_0 \in [a, b]$ such that $f(x_0) = c$.*

Proof. Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Without loss of generality, assume $f(a) < f(b)$. We choose an arbitrary $c \in \mathbb{R}$ such that $f(a) \leq c \leq f(b)$ and define the set:

$$X := \{t \in [a, b] \mid f(t) \leq c\}$$

Observe that $X \neq \emptyset$ (since $a \in X$ as $f(a) \leq c$) and X is bounded above by b. Thus, by the completeness axiom of the real numbers, the supremum $s := \sup X$ exists, and clearly $s \in [a, b]$.

By the definition of the supremum, for all $n \in \mathbb{N}$, there exists $t_n \in X \cap [s - \frac{1}{n}, s]$. Because $t_n \to s$ as $n \to \infty$ and f is continuous, we have $\lim_{n \to \infty} f(t_n) = f(s)$. Since $t_n \in X$, we know $f(t_n) \leq c$ for all $n \in \mathbb{N}$. By the limit-preserving properties of inequalities, it follows that $f(s) \leq c$.

Now, assume for the sake of contradiction that $f(s) < c$. Since $f(s) < c \leq f(b)$, we know $s \neq b$, which implies $s < b$.

Let $\epsilon = c - f(s) > 0$. By the continuity of f at s, there exists a $\delta > 0$ such that for all $y \in [a, b]$:

$$|y - s| < \delta \implies |f(y) - f(s)| < \epsilon$$

This implies $f(y) < f(s) + \epsilon = c$.

Now, define $y := \min(b, s + \frac{\delta}{2})$. Since $s < b$ and $\delta > 0$, we strictly have $y > s$. Furthermore, $y \in [a, b]$ and $|y - s| = \frac{\delta}{2} < \delta$. Therefore, by our continuity implication, $f(y) < c$, which means $y \in X$.

However, we have found an element $y \in X$ such that $y > s$, which directly contradicts the fact that s is an upper bound (the supremum) of X.

Thus, our assumption that $f(s) < c$ must be false. Since we have already established that $f(s) \leq c$, we conclude that $f(s) = c$. $\square$

Definition 4.8. *A bounded and closed interval is called* ***compact****.*

Lemma 4.9. *Let $[a, b]$ be a compact interval and let $(x_n)_{n\in\mathbb{N}} \subset [a, b]$ be a sequence. Then there exists a subsequence $(x_{n_k})_{k\in\mathbb{N}}$ such that*

$$\lim_{k\to\infty} x_{n_k} = x_0 \text{ for some } x_0 \in [a, b].$$

Definition 4.10. *Let $f : D \subset \mathbb{R} \to \mathbb{R}$ be a function*

- *If for $x_0 \in D$ it holds that $f(x_0) \geq f(x) \forall x \in D$, then f has a local maximum at x_0.*
- *If for $x_0 \in D$ it holds that $f(x_0) \leq f(x) \forall x \in D$, then f has a local minimum at x_0.*
- *Local maxima and minima are called local extrema.*

Theorem 4.11. *Let $f : I \subset \mathbb{R} \to \mathbb{R}$ be a continous function and I be a compact interval. Then f is bounded and attains its maximum and minimum value on I. Hence*

$$\exists x_{min}, x_{max} \in I \text{ such that } f(x_{min}) \leq f(x) \leq f(x_{max}) \forall x \in I$$

5 Differential calculus

Definition 5.1. *The* ***average rate of change*** *of a function f in the domain between a and $a+h$ is defined as*

$$\frac{\Delta y}{\Delta x} = \frac{f(a+h) - f(a)}{h}$$

and also called the differential quotient.

Definition 5.2. *The* ***derivative*** *f' of a function f at the coordinate a is defined as*

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = lim_{b \to a} \frac{f(b) - f(a)}{b - a} = \frac{df}{dx}|_{x=a} = f'(a)$$

Remark 5.3. *The derivative $f'(a)$ can geometrically interpreted as the incline/decline of the graph of f at the point $(a, f(a))$.*

Definition 5.4. *If for a function f at the cooridnate a the derivative $\exists f'(a)$ then f is called* ***differentiable*** *at a. If f is differentiable for every $a \in \mathbb{D}$ then f is called differentiable.*

Example 5.5. *let $f(x) = |x|$ we can see that for $a = 0$ there is no derivative (we have a sharp change in "direction")*

Remark 5.6. *We can generally expect that there are no derivatives at undefined points $a \notin \mathbb{D}$.*

Lemma 5.7. *If f is differentiable at x_0 then f is continous at x_0.*

Proof. Assume f differentiable then

$$\lim_{x \to x_0} f(x) = \lim_{h \to 0} f(x_0 + h) = \lim_{h \to 0} f(x_0 + h) - f(x_0) + f(x_0)$$

$$= f(x_0) + \lim_{h \to 0} f(x_0 + h) - f(x_0) = f(x_0) + \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h} \cdot h$$

$$= f(x_0) + \lim_{h \to 0} h \cdot \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

Observe that the last term exists by the assumption hence

$$= f(x_0) + 0 \cdot f'(x_0) = f(x_0)$$

□

Definition 5.8. *If the limit*

$$\lim_{h>0\to 0} \frac{f(a+h)-f(a)}{h}$$

existst then we call it the ***right-sided derivative*** *(or the left-sided derivative analogue)*

Definition 5.9. *For a function f we can iteratively define higher derivatives for $n \in \mathbb{N}_0$*

$$f^(0) = f, f^(1) = f', f^(2) = f'', \dots f^(n+1) = (f^(n))'$$

if $f^(n)$ exists we call it n-times differentiable.

Definition 5.10. *if $f^(n)$ is continous we call f n-times continous differentiable. The set of all n-times continous differentiable functions on D is denoted as $C^n(D)$*

Definition 5.11. *We call $C^\infty(D) := \cap_{n=0}^{\infty} C^n(D)$ the set of smooth functions.*

5.1 Derivation rules

Theorem 5.12. *For functions $f(x)$ we have*

- $f(x) = ax^p \implies f'(x) = apx^{p-1}$
- $f(x) = a^x \implies f'(x) = ln(a) \cdot a^x$
- $f(x) = sin(x) \implies f'(x) = cos(x)$
- $f(x) = cos(x) \implies f'(x) = -sin(x)$
- $f(x) = tan(x) \implies f'(x) = \frac{1}{cos^2(x)} = 1 + tan^2(x)$
- $f(x) = sinh(x) \implies f'(x) = cosh(x)$
- $f(x) = cosh(x) \implies f'(x) = sinh(x)$
- $f(x) = ln(x) \implies f'(x) = \frac{1}{x}$
- $f(x) = arcsin(x) \implies f'(x) = \frac{1}{\sqrt{1-x^2}}$
- $f(x) = arccos(x) \implies f'(x) = \frac{-1}{\sqrt{1-x^2}}$
- $f(x) = arctan(x) \implies f'(x) = \frac{1}{1+x^2}$

Theorem 5.13. ***Sum Rule:*** *For the n-th derivative of a sum:*

$$(f+g)^{(n)} = f^{(n)} + g^{(n)}$$

Proof. This can be proved inductively by applying the definition of the derivative n times.

$$(f+g)'(x) = lim_{h\to 0}\frac{f(x+h)+g(x+h)-f(x)-g(x)}{h}$$

$$= lim_{h\to 0}\frac{f(x+h)-f(x)}{h} + lim_{h\to 0}\frac{g(x+h)-g(x)}{h}$$

$$= f'(x) + g'(x)$$

For $n = 2$ we have

$$(f+g)''(x) = (f'+g')'(x) = f''(x) + g''(x)$$

and so on. □

Theorem 5.14. ***General Leibniz Rule (Product):*** *The n-th derivative of a product is given by:*

$$(f\cdot g)^{(n)} = \sum_{k=0}^{n}\binom{n}{k} f^{(n-k)} g^{(k)}$$

Proof. This is proved analogously to the sum rule.

$$(f\cdot g)'(x) = lim_{h\to 0}\frac{f(x+h)g(x+h)-f(x)g(x)}{h}$$

$$= lim_{h\to 0}\frac{f(x+h)g(x+h)-f(x)g(x+h)+f(x)g(x+h)-f(x)g(x)}{h}$$

$$= lim_{h\to 0}\frac{g(x+h)(f(x+h)-f(x))}{h} + lim_{h\to 0}\frac{f(x)(g(x+h)-g(x))}{h}$$

$$= g(x)f'(x) + f(x)g'(x)$$

For $n = 2$ we have

$$(f\cdot g)''(x) = (f'g+fg')'(x) = f''g + f'g' + f'g' + fg'' = f''g + 2f'g' + fg''$$

and so on. □

Theorem 5.15. ***Chain Rule:*** *For the composition of functions:*

$$(f\circ g)'(x) = f'(g(x))\cdot g'(x)$$

Proof. Observe that

$$g(y) = g(y_0) + g'(y_0)(y - y_0) + g(y) - g(y_0) - g'(y_0)(y - y_0)$$

with the first two summands being a linear (tangential) approximation of $g(y)$ near y_0

$$g(y) \approx g(y_0) + g'(y_0)(y - y_0)$$

By taking out $(y - y_0)$ of the last two summands in the original equation we get:

$$g(y) = g(y_0) + g'(y_0)(y - y_0) + \left[\frac{g(y) - g(y_0)}{y - y_0} - g'(y_0)\right](y - y_0)$$

We define $\epsilon = \frac{g(y)-g(y_0)}{y-y_0} - g'(y_0)$ and get

$$\lim_{x \to x_0} \frac{g(f(x)) - g(f(x_0))}{x - x_0}$$

$$= \lim_{x \to x_0} \left(f(f(x_0)) + g'(f(x_0))[f(x) - f(x_0)] + \epsilon f(x)[f(x) - f(x_0)] - g(f(x_0))\right) \frac{1}{x - x_0}$$

$$= g'(y_0)f'(x_0)$$

□

Theorem 5.16. ***Quotient Rule:*** *For $g(x) \neq 0$:*

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

Remark 5.17. *This follows directly from the product rule.*

Theorem 5.18. ***Inverse Functions:*** *If f is differentiable and injective:*

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

5.2 Local extrema

Theorem 5.19. *Let $I \subset \mathbb{R}$ be a an intervall, $f : I \to \mathbb{R}$, and let x_0 be a local Extrema of f. Then one of the following holds:*

- *x_0 is an endpoint of I*
- *f is not differentiable at x_0*
- *f is differentiable at x_0 and $f'(x_0) = 0$*

Remark 5.20. *From $f'(x_0) = 0$ we can not conclude that x_0 is an extremum.*